

EDUCATION

University of California, Davis
B.S. Electrical & Computer Engineering (EEC)
Expected Graduation: 2027

SKILLS

- C, C++, Python, MATLAB, Verilog, JavaScript
- **FPGA design**, digital logic design, **embedded systems**, microcontrollers, hardware debugging, interrupt programming, **UART**, **SPI**, **I2C**, serial communication
- **RISC-V Assembly**, **Hardware Acceleration**, Parallel Computing, System Integration
- **Machine Learning**, Model Training, Neural Networks, Data Analysis, Model Evaluation
- **Signal Processing Fundamentals**, Audio Signal Processing, **Live Sound Engineering**, **DJ'ing**
- **Film production**: Video (**PR**)/picture(**PS**), editing,
- **Microsoft Office Suite** (Excel, Word, PowerPoint)

ENGINEERING PROJECTS

- **Matrix Multiplication Hardware Accelerator** – Designed FPGA-based accelerator using Verilog implementing multiply-accumulate architecture and parallel computation.
- **IoT Cloud Monitoring System** – Built embedded device integrating sensors with AWS cloud services using REST APIs for real-time monitoring.
- **Jelly Bean Counting Scale** – Developed load-cell measurement system with instrumentation amplifier and MATLAB GUI for real-time data acquisition and estimation.
- **GEMINI Senior Design Project** – Contributed to AI-enabled sensing and modeling system supporting climate-resilient agricultural analysis.
- **Applied Machine Learning Project** – Implemented ML models including training pipelines, performance evaluation, and data-driven prediction methods.
- **ATM System** – Developed C++ application simulating banking operations including deposits, withdrawals, and account management.
- **IR Remote System** – Designed interrupt-driven embedded firmware decoding IR signals and transmitting data over UART for real-time device control.

HEAVEN BERHE

HARDWARE & SOFTWARE SYSTEMS | MACHINE LEARNING

PROFILE



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EEC student at UC Davis specializing in **hardware-software systems, embedded development, software engineering, and applied machine learning**. Hands-on experience in FPGA design, embedded firmware, and building real-world engineering systems involving sensing, cloud connectivity, and intelligent computing.

WORK EXPERIENCE

Tutor, Self-employed

2022 - PRESENT

- Facilitated engaging tutoring sessions in various subjects, including math and science.
- Explained complex concepts using clear problem-solving strategies to improve student performance and confidence.
- Tracked learning progress and adjusted teaching methods based on student needs.

Coding Intern, Dropbox

JUN 2020 - MAR 2021

- Developed and enhanced web-based application features using JavaScript, HTML, and front-end scripting techniques, improving user interaction and performance.
- • Implemented dynamic UI components and optimized client-side functionality to increase responsiveness and usability.
- • Debugged and tested application modules, identifying and resolving performance bottlenecks and functional issues.
- • Collaborated with cross-functional engineering teams to deliver production-ready features aligned with technical requirements and project timelines.

Intern, NASA

NOV 2022 - JUL 2023

- Collaborated with NASA to distribute resources and disseminate knowledge about solar systems to underprivileged communities, contributing to educational outreach initiatives.

Student Assistant, Berkley City College(BCC)

SEP 2022 - MAY 2023

- Assisted students with the process of transferring to a 4-year University, providing guidance and support throughout
- Offered academic and life support to students in need, fostering a supportive and inclusive environment.
- Updated websites, created flyers, and designed posters to promote events and programs
- Digitized and organized student files, ensuring easy access and retrieval of information

Co-Founder & President|Ambassador & Translator(OIHS)

JUN 2020 - MAR 2021

- Co-founded and led a 100+ student organization, organizing school activities and representing student voices both online and in person.
- Supported new and current students by connecting them to academic, social, and community resources, including clubs, student government, and support programs.
- Increased student engagement and family involvement by 60% through outreach and communication initiatives.
- Spearheaded the design and launch of the organization's first yearbook, documenting student achievements and community events.

Social Media Coordinator & Multimedia Specialist/ DJ

2022 - PRESENT

- Manage YouTube, Facebook, and Instagram content
- Film, edit, and produce video and multimedia content
- Operate cameras and support live media production
- Mix live sound and DJ events using professional audio equipment

LEADERSHIP & RECOGNITION

- Student Ambassador certificate (OIHS) 
- Published Author, The Bare Life Review (2021) 